

Homework # 3

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Crypto Campaigns: Co-Participation Network Analysis

Original idea: A social analysis of event co-participation could be formed, where nodes represent campaign participants and edges between them represent participation in the same campaign, a similar analysis could be carried out for specified comment activity in the same forum threads. The resulting network could be analyzed for degree distribution, centrality, clustering, k-shell analysis, and more. From this analysis participation or activity patterns could be determined and potentially contrasted with other social networks (non-crypto related) for comparison.

Exploratory Data Analysis

An undirected co-participation network was constructed from the Bitcointalk Bounties (Altcoins) dataset, where nodes represent users and edges represent shared participation in bounty campaigns. To reduce noise, a minimum threshold of ≥ 3 shared campaigns was applied. The tables below report summary statistics of the resulting filtered network.

General Characteristics

Table 1 reports basic statistics of the dataset and the resulting network after applying the co-participation threshold.

Table 1: General Statistics from Co-Participation Data in `comments.participation.tsv`

Parameter	Value
Unique Users	$\approx 74.5\text{K}$
Unique Campaigns	$\approx 5.6\text{K}$
Campaigns with ≤ 500 Users	$\approx 4.5\text{K}$
Total Co-Participation Instances	$\approx 83.8\text{M}$
Co-Participation Weight Threshold	$\#campaigns \geq 3$
Network Nodes (after filtering)	26,032
Network Edges (after filtering)	6,719,776
Isolated Nodes Removed (degree = 0)	$\approx 48.5\text{K}$
Type of Graph	Undirected

Table 2 summarizes the degree distribution of the filtered co-participation network.

Table 2: Degree Statistics from Co-Participation Network ($\#campaigns \geq 3$)

Parameter	Value (rounded)
Min Degree	1
Max Degree	13,057
Avg Degree	516.27
Median Degree	182
Standard Deviation of Degree	805.53

Table 3 reports the size of the largest connected component in the filtered network.

Table 3: Largest Connected Component ($\#campaigns \geq 3$)

Parameter	Value (rounded)
Nodes	25,963
Edges	6,719,669
Percentage of Total Nodes	99.7%

Node Centrality

Degree centrality was computed for all nodes. Table 4 lists the five users with the highest degree values.

Table 4: Top 5 Nodes by Degree Centrality ($\#campaigns \geq 3$)

Username	User ID	Degree
LastKiss	14413	13,057
rudilo	4065	12,132
siena23	17656	10,212
banyubiru	19270	9,548
jendervan	19786	9,074

Closeness centrality was computed on the largest connected component for the top 100 nodes by degree. Table 5 reports the top five users by closeness.

Table 5: Top 5 Nodes by Closeness Centrality (Among Top 100 by Degree, Largest Connected Component)

Username	User ID	Closeness
LastKiss	14413	0.6640
rudilo	4065	0.6483
siena23	17656	0.6144
banyubiru	19270	s0.6059
jendervan	19786	0.5990

PageRank scores were computed on the largest connected component. Table 6 lists the five users with the highest PageRank values.

The user with the highest degree centrality, PageRank, and closeness centrality (among top 100 nodes) is LastKiss, a bounty participant ranked as Sr. Member on Bitcointalk with 250 merits, 8,792 posts, and active participation in comments, events, and subforums.

Table 6: Top 5 Nodes by PageRank (Largest Connected Component)

Username	User ID	PageRank
LastKiss	14413	0.001342
tamango	21578	0.001183
rudilo	4065	0.001000
gsmbimol	25078	0.000973
nilbtc4555	11538	0.000893

Path Length and Diameter

Table 7 reports path length metrics for the largest connected component.

Table 7: Path Length and Diameter of Largest Connected Component (25,963 Nodes)

Metric	Value
Effective Diameter	2.76
Diameter	6
Average Shortest Path	2.37

K-shell Decomposition

The k-shell decomposition is presented in Figure 1, which shows the structure of coreness levels in the network.

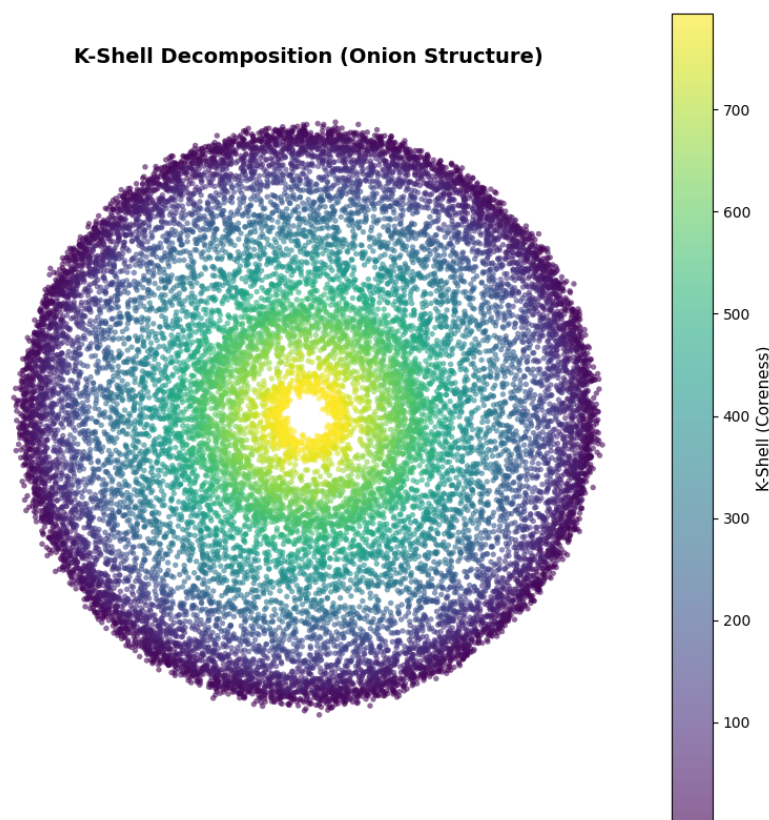


Figure 1: K-shell decomposition of the co-participation network.

Clustering Coefficient

Table 8 reports the global clustering coefficient and total number of triangles in the network.

Table 8: Clustering Coefficient and Triangles

Metric	Value
Clustering Coefficient	0.73
Number of Triangles	1,404M

Power-law Fit

The degree distribution of the network was analyzed for power-law characteristics. Figure 2 shows the degree distribution in linear scale, while Figure 3 presents it in log-log scale to assess power-law behavior.

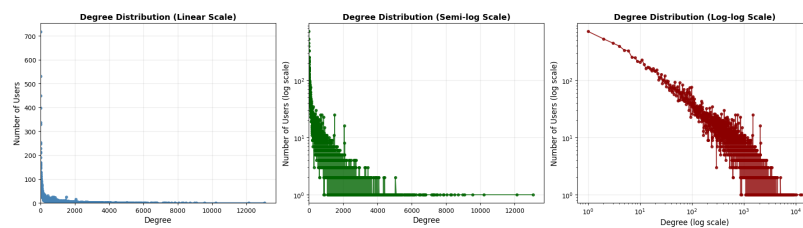


Figure 2: Degree distribution of the co-participation network in linear scale.

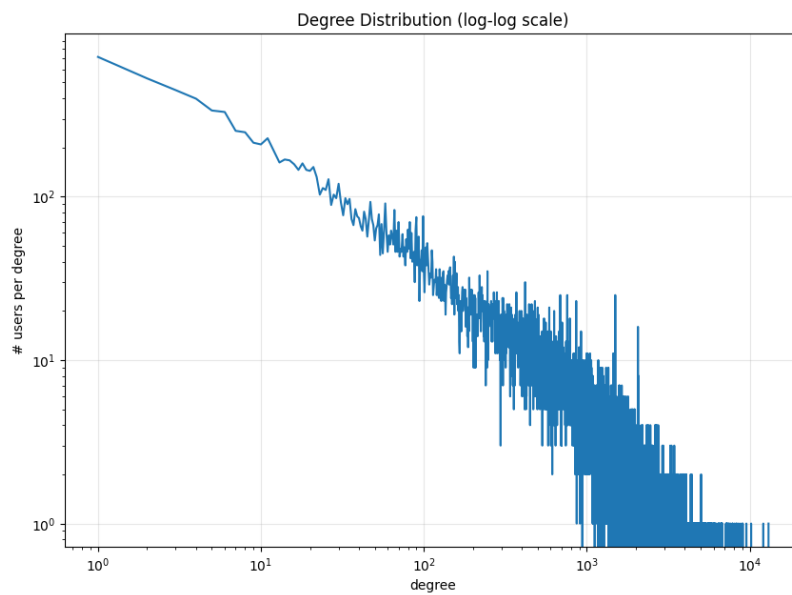


Figure 3: Degree distribution in log-log scale, revealing power-law-like behavior in the tail.

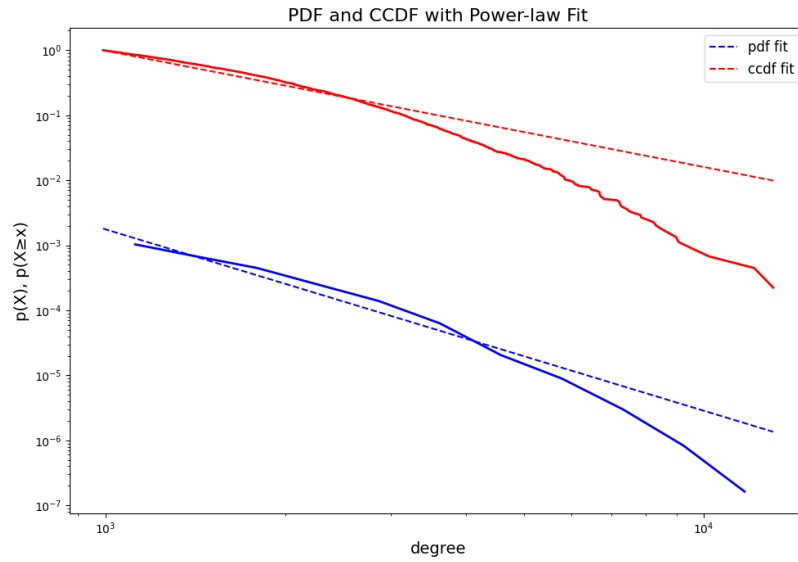


Figure 4: Probability density function (PDF) and complementary cumulative distribution function (CCDF) of the degree distribution.

Network Comparison

Table 9 compares key statistics of the empirical network with several generative and null models of similar size.

Table 9: Comparison of empirical network statistics vs. generative/null models. Ratios are relative to the real network.

Model	Clustering C	Skewness	Edges
Real Network	0.7306	3.306	6,719,776
Preferential Attachment	0.1019 (0.14×)	3.086	13,299,126 (1.98×)
Configuration Model	0.1652 (0.23×)	2.465	6,074,096 (0.90×)
Erdős-Rényi	0.0198 (0.03×)	0.034	6,719,776 (1.00×)
Node Rewiring	0.1652 (0.23×)	2.465	6,074,096 (0.90×)

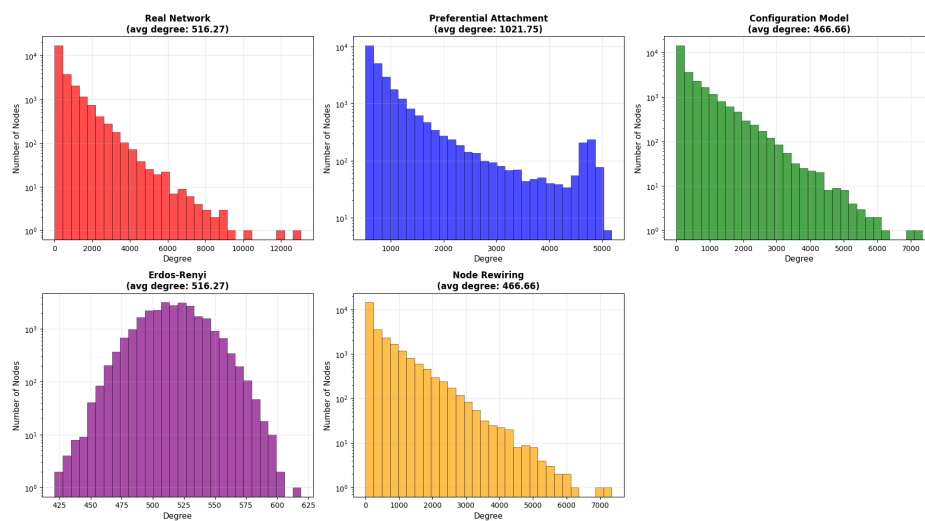


Figure 5: Visual comparison of network statistics across empirical and generative models, highlighting differences in clustering and connectivity patterns.

Discussion

- All centrality measures (degree, closeness, PageRank) identified a similar set of influential users (e.g., LastKiss, rudilo). These users do not only have high degree but also play a central role in connecting others, which may suggest they are some sort of coordinators in the bounty environment.
- We can observe that degree and PageRank are not identical. E.g. tamango has a high PageRank despite a lower degree than some others, suggesting that being connected to influential users matters.
- The average shortest path length (2.37) and effective diameter (2.76) shows a small-world structure, meaning that information and collaboration can spread fast through the network.
- The small maximum diameter (6) indicates a highly interconnected network where most users are only a few steps away from each other.
- The clustering coefficient (0.73) is very high, and tells us that users who collaborate together tend to form close groups. This suggests strong community structure, where users repeatedly work together across multiple campaigns.
- The large amount of triangles (1.4B) additionally supports that users who collaborate together tend to form groups and work together repeatedly. This is consistent with the high clustering coefficient.
- The k-shell structure of the co-participation network most closely resembles the “Actors” network seen in class, where there is a single core and multiple dense shells surrounding it. This suggests that participants collaborate in nested communities.
- $C = 0.73 \gg \text{PA, CM, ER} \Rightarrow$ This tells us that people who share a campaign tend to also share other campaigns with the same people. This talks about the presence of strong local/community structure in the network, which is not captured by PA or CM.
- Degree skew (≈ 3.31) This is a highly skewed distribution, indicating that while most users have low degree (participate in few campaigns), a small number of users are extremely well-connected (participate in many campaigns). This suggests a power-law or scale-free structure, which is consistent with the presence of hubs.
- Rewiring and configuration models have similar clustering and skewness, but the real network has much higher clustering than either model, indicating that the real network has more local structure than expected by degree distribution alone (if we shuffle the connections but keep everyone equally active, the groups disappear).
- Network structure reflects hubs and strong community organization.
- Overall, the results suggest that the Bitcointalk bounty ecosystem is organized around a dense core of highly active users who repeatedly collaborate across campaigns, forming stable communities that dominate participation and connectivity patterns.

Project Proposal Refinement

Now that general analysis was performed, the original idea appears underspecified and superficial. A refined idea would be to focus on the core-periphery structure of the Bitcointalk bounty co-participation network. The updated goal would be:

- Distinguish between a small, dense core of highly active users and a large peripheral group with limited participation.
- Analyze how core users differ from peripheral users in terms of degree, clustering, and network centrality.
- Examine whether repeated collaboration is concentrated among core users across multiple campaigns.
- Assess how much the core contributes to the overall connectivity and cohesion of the network.
- Compare core-periphery structure with other models to understand if it is a unique feature of this ecosystem or a common pattern in collaborative networks.

In general, the goal is to understand who dominates participation and how network structure reflects inequality in collaboration.